Queen Mary
University of London

EBU5376
Digital Signal Processing

DSP Systems – Stability

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Outline

- Transfer Function
  - Pole-Zero Plot
  - Stability

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Transfer Function – Poles and Zeros

- We can analyse the stability of a LTI system by identifying the poles and zeros of the transfer function $H(z) = \frac{N(z)}{D(z)}$

- **Zeros**
  - Zeros are points in the complex plane where the transfer function becomes zero.
  - They are associated with the roots of the numerator polynomial in the transfer function, i.e., $N(z) = 0$
  - The location of zeros influences the system's response and the frequencies at which it attenuates or amplifies signals.

Transfer Function – Poles and Zeros

- We can analyse the stability of a LTI system by identifying the poles and zeros of the transfer function $H(z) = \frac{N(z)}{D(z)}$

- **Poles**
  - Poles are points in the complex plane where the transfer function becomes infinite.
  - They are associated with the roots of the denominator polynomial in the transfer function, i.e., $D(z) = 0$
  - The location of poles affects the stability and dynamics of the system.

Transfer Function – Poles and Zeros

- <u>Example</u>: Find the poles and zeros of the transfer function:

$$H(z) = \frac{z+2}{z^2 + \frac{1}{4}}$$

Then, plot the poles and zeros on the same coordinate system.

- Use ‘o’ as the symbol for zeros
- Use ‘x’ as the symbol for poles
- Draw the unit circle ($|z| = 1$)

Transfer Function – Poles and Zeros

- Example: Find the poles and zeros of the transfer function:

$$H(z) = \frac{z+2}{z^2 + \frac{1}{4}}$$

The zeros of the system can be obtained by equating the numerator of the transfer function to zero:

$$z + 2 = 0 \Rightarrow z = -2$$

The poles of the system can be obtained by equating the denominator of the transfer function to zero:

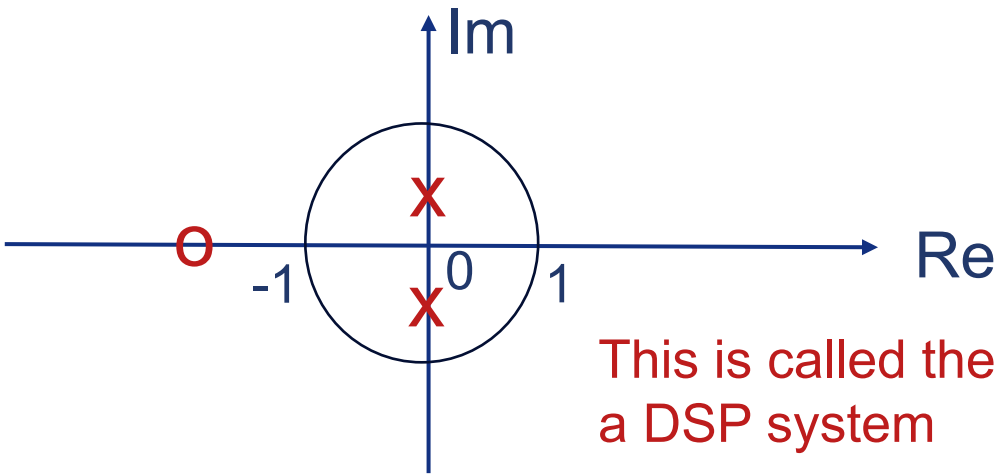
$$z^2 + \frac{1}{4} = 0$$

$$\left(z + \frac{1}{2}j\right)\left(z - \frac{1}{2}j\right) = 0 \Rightarrow z = -\frac{1}{2}j, or\ z = \frac{1}{2}j$$

Transfer Function – Poles and Zeros

- Example: Find the poles and zeros of the transfer function:

$$H(z) = \frac{z+2}{z^2 + \frac{1}{4}}$$



This is called the **pole-zero plot** of a DSP system

Transfer Function – Poles and Zeros

- Example: Draw the pole-zero plot of the following system:

$$y[n] - 2y[n-1] = x[n-1] + x[n-2]$$

Transfer Function – Poles and Zeros

- Example: Draw the pole-zero plot of the following system:

$$y[n] - 2y[n-1] = x[n-1] + x[n-2]$$

First, we need to derive the transfer function:

$$Y(z)(1 - 2z^{-1}) = X(z)(z^{-1} + z^{-2})$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{z^{-1} + z^{-2}}{1 - 2z^{-1}} = \frac{z+1}{z^2 - 2z}$$

Then, the zeros are calculated as:

$$z + 1 = 0$$

$$\Rightarrow z = -1$$

The poles is calculated as

$$z^2 - 2z = 0$$

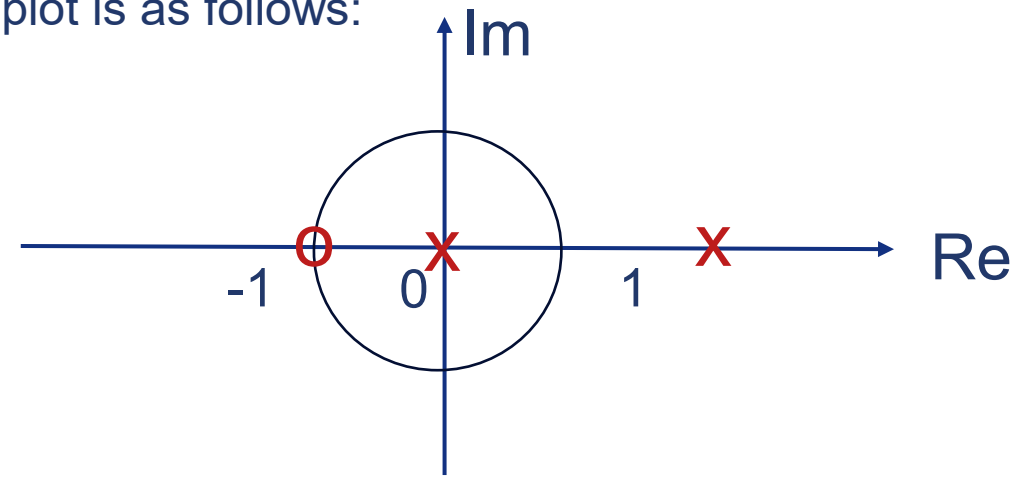
$$\Rightarrow z = 2, z = 0$$

Transfer Function – Poles and Zeros

- Example: Draw the pole-zero plot of the following system:

$$y[n] - 2y[n-1] = x[n-1] + x[n-2]$$

The zero is $z = -1$. The poles are $z = 2, z = 0$.
The pole-zero plot is as follows:



Transfer Function – Poles and Zeros Summary

- A pole-zero plot is a graphical representation of the locations of the poles and zeros of the transfer function of a system in the complex plane.

- **System Characteristics**: The pattern and distribution of poles and zeros provide insights into the system's behavior and stability. The interaction between poles and zeros determines how the system responds to different input frequencies.

Outline

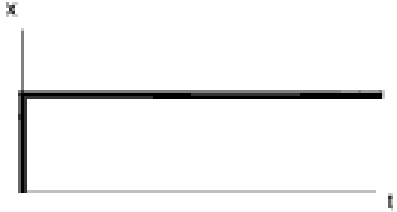
- Transfer Function
  - Pole-Zero Plot
  - Stability

Stability

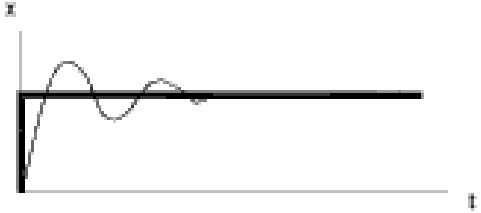
- This module studies asymptotic stability, not BIBO stability.
- Asymptotic stability refers to the system's ability to return to an equilibrium state after a disturbance has ceased.
- In terms of system response:
  - **Stable systems** will return to equilibrium or a steady-state after being disturbed.
  - **Unstable systems** will diverge away from an equilibrium, often growing without bound in response to any disturbance.
  - **Marginally stable** systems will neither return to equilibrium nor diverge but will either continue oscillating indefinitely with a constant amplitude or remain at a fixed constant output for any input.

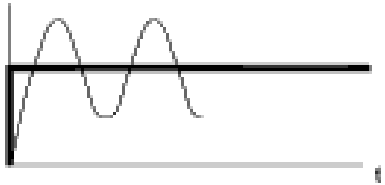
Stability

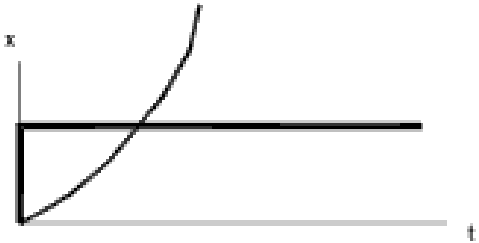
- Given the system responses, determine the stability of the following 5 systems.

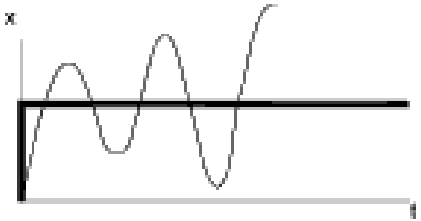


INPUT









Stability

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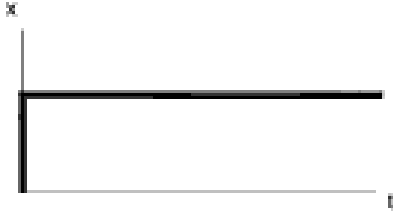
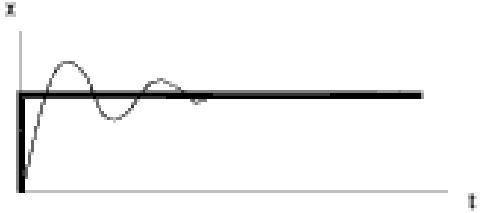
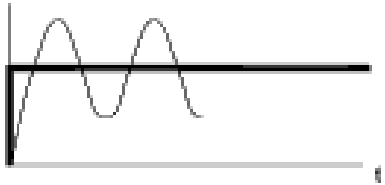


INPUT

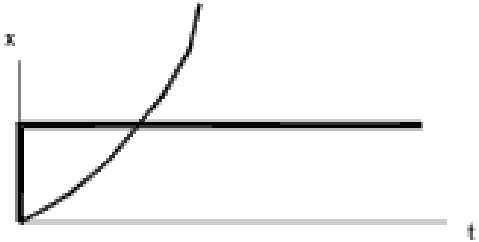


STABLE

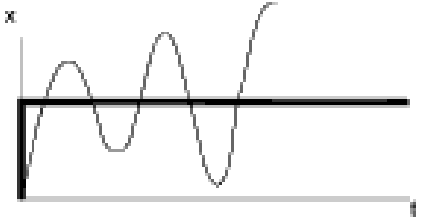
STABLE



MARGINALLY STABLE



UNSTABLE



UNSTABLE

Stable Systems

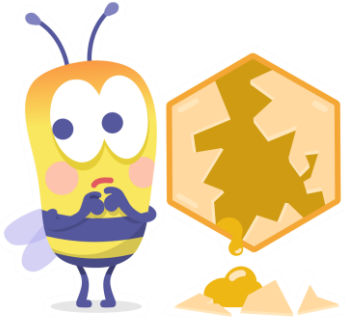
- **Stable systems** are useful because the outputs are always bounded for any bounded inputs

- **Stable systems** are easier to control and design
  - They exhibit *predictable and repeatable behavior* that can be analyzed and modeled using mathematical tools.
  - This makes it *easier to achieve desired system performance*, meet design specifications, and optimize system performance.

Unstable Systems

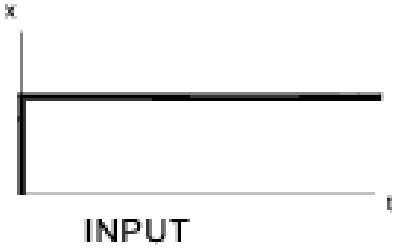
- **Unstable systems** are not useful because the output can grow without bound.
  - E.g., Microphone feedback loop

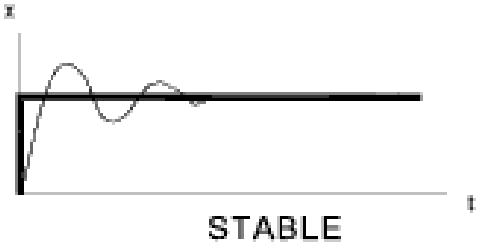
- Unbounded output can cause *damage to equipment, introduce errors into a system, or render the system unusable*

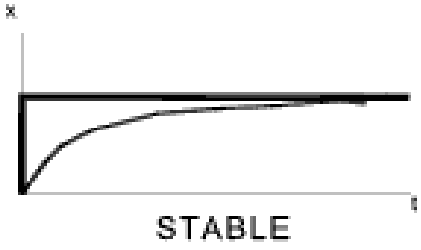


Marginally Stable Systems

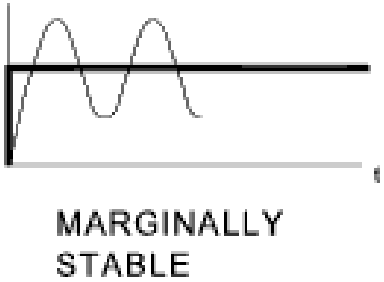
- We also want to avoid **marginally stable systems** because the system is on the edge of stability.
- Any small changes can cause the system to become unstable.
- Marginally stable systems can be useful in certain applications
  - in a musical instrument to produce a sustained tone
  - in a mechanical system to produce a controlled vibration or oscillation.

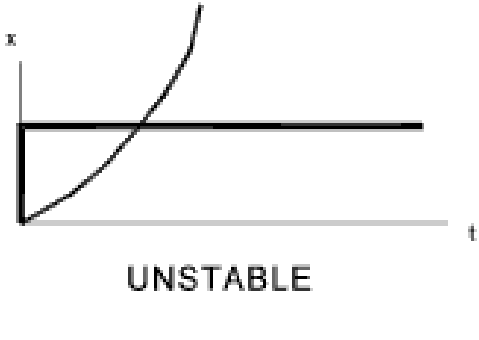


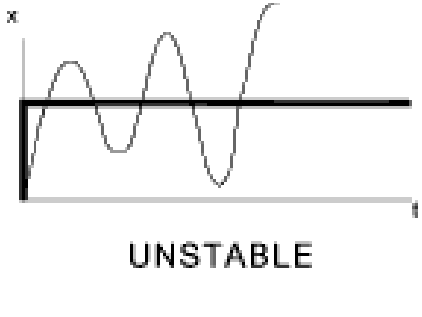




How to Determine Stability?







Stability

- Try to describe the stability of the following systems.

1. $y[n] = 2x[n] + x[n-1]$

2. $y[n] = y[n-1] + x[n]$

3. $y[n] = u[n]$

Stability

- Try to describe the stability of the following systems.

1. $y[n] = 2x[n] + x[n-1]$ Stable
2. $y[n] = y[n-1] + x[n]$ Marginally Stable
3. $y[n] = u[n]$ Marginally Stable

Stability

- Try to describe the stability of the following systems.

1. $y[n] = x[n] + x[n-1]$

2. $y[n] = 0.1y[n-1] + x[n] + x[n-1]$

3. $y[n] = 0.9y[n-1] + x[n] + x[n-1]$

4. $y[n] = 1.1\, y[n-1] + x[n] + x[n-1]$

Stability

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2. $y[n] = 0.1y[n-1] + x[n] + x[n-1]$ Stable
3. $y[n] = 0.9y[n-1] + x[n] + x[n-1]$ Stable
4. $y[n] = 1.1\,y[n-1] + x[n] + x[n-1]$ Unstable

Stability

- Draw the pole-zero plots of the following system

1. $y[n] = x[n] + x[n-1]$

2. $y[n] = 0.1y[n-1] + x[n] + x[n-1]$

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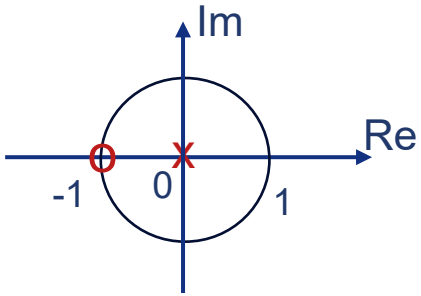

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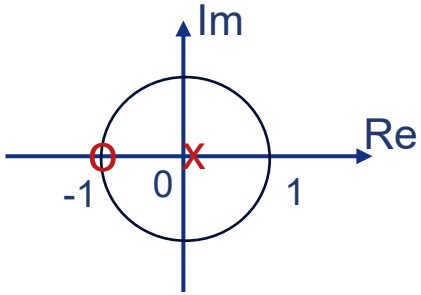

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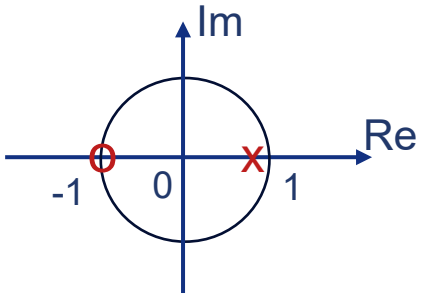
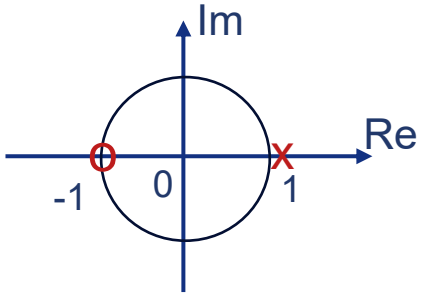


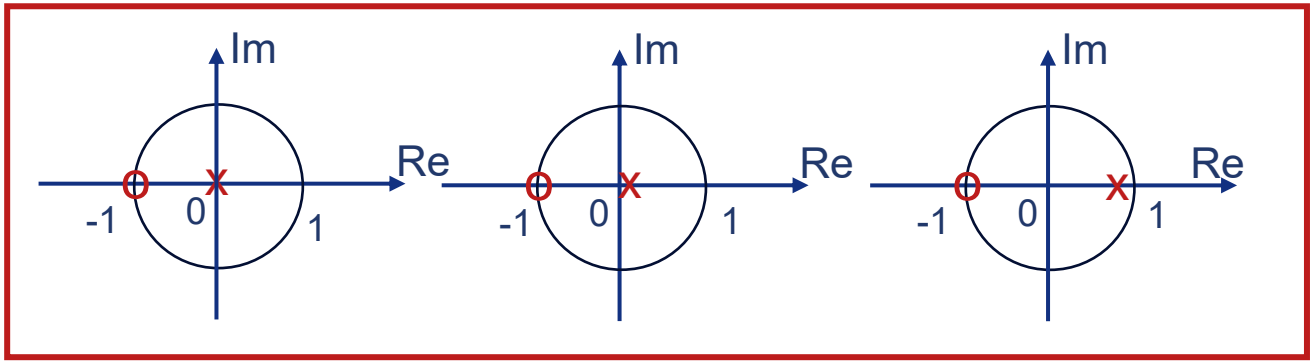
4. $y[n] = 1.1\, y[n-1] + x[n] + x[n-1]$



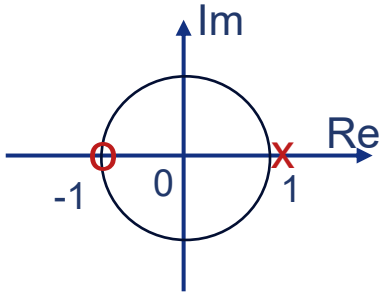
Stability

- Observations:
  - As absolute values of the coefficients of past outputs increases, the poles approaches the unit circle.
  - If these coefficients are greater than one (or less than -1), the system response will tend to infinity, hence the system is unstable

Stable systems



Unstable system



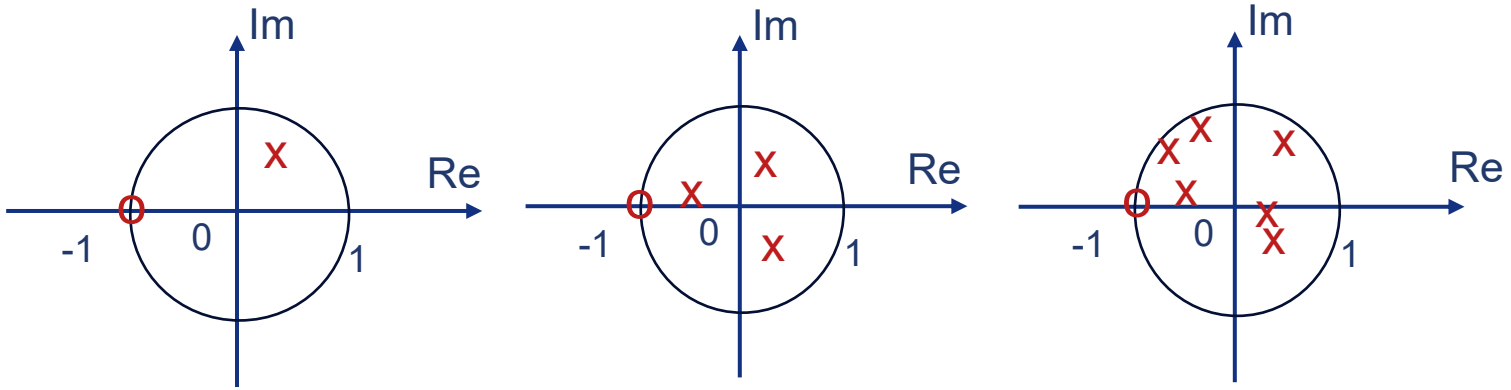
Stability - Conclusion

- For a system to be **stable**, all poles of the system's transfer function must be inside the unit circle in the z-plane. The unit circle represents stability, and poles inside it ensure that the system's response does not grow without bound.
- Zeros can be located anywhere in the z-plane, as they don't affect stability

❖ The above conclusions assumes that the system is causal.

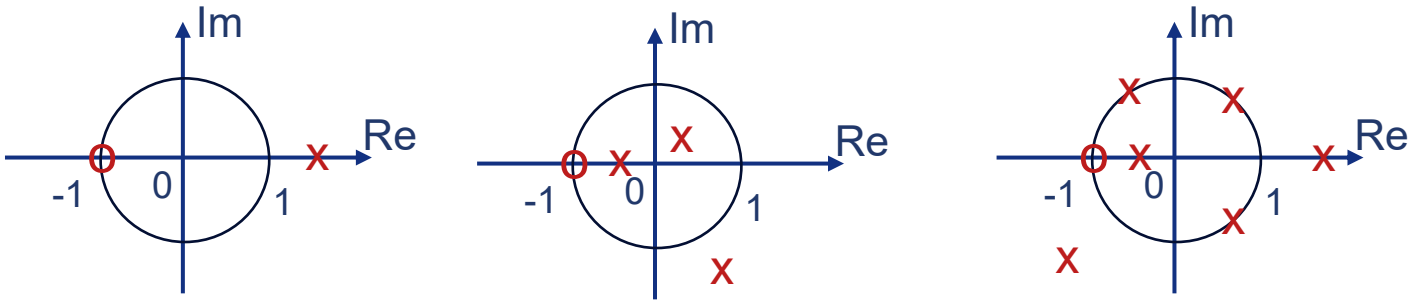
Stability

➢ The following are pole-zero plots of **stable systems**:



Stability

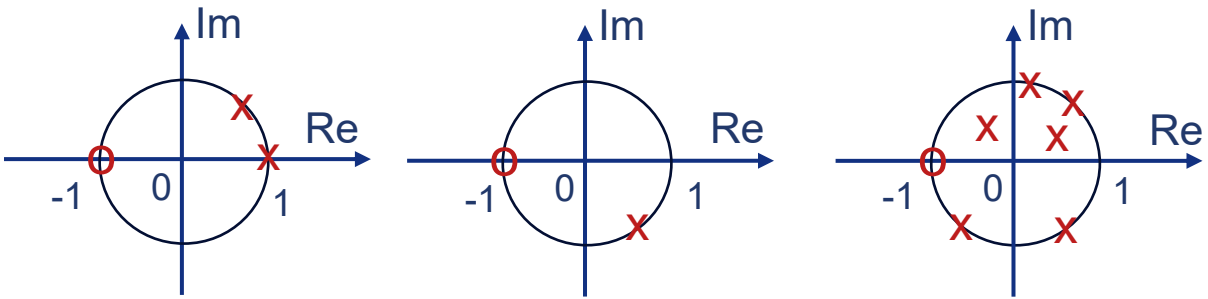
- If at least one of the poles is outside the unit circle, the system becomes **unstable**
- The following are pole-zero plots of unstable systems:



Stability

- If there are poles located on the unit circle, we need to further check if the poles are located at the same place.
  1. If all poles on the unit circle are located at different places, then the system is **marginally stable**

     The following are pole-zero plots of marginally stable systems:

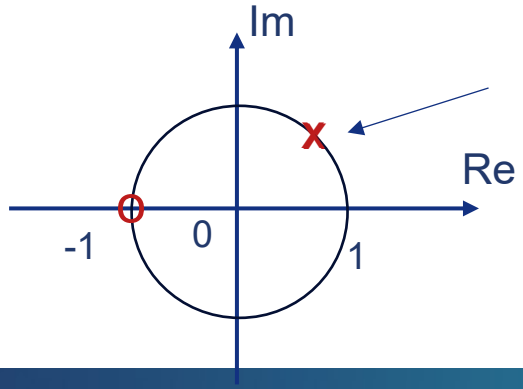


I.e., the poles on the unit circle are distinct: $|z_i| = |z_j| = 1$, but $z_i \neq z_j$

Stability

- If there are poles located on the unit circle, we need to further check if the poles are located at the same place.
  2. If at least one pair of the poles are located at the same place on the unit circle, then the system is **unstable**

The following is the pole-zero plot of an unstable system:



Duplicated poles on the unit circle, i.e., $|z_1| = |z_2| = 1$ and $z_1 = z_2$

Why Do Repeated Poles on the Unit Circle Cause Instability?

Consider a system with a transfer function $H(z)$ that has a repeated pole at $z = e^{j\omega_0}$

The partial fraction expansion of $H(z)$ will include terms of the form:

$$\frac{A}{(z - e^{j\omega_0})^k}$$

Where $k \geq 2$ (indicating a repeated pole)

- The inverse z-transform of such terms results in time-domain signals that grow polynomially with n.
- For example, if $k = 2$:

$$Z^{-1}\left(\frac{A}{(z - e^{j\omega_0})^k}\right) = Ane^{j\omega_0 n}u[n]$$

Here the term $ne^{j\omega_0 n}$ grows linearly with n, causing the impulse response to grow unbounded

For higher values of k, the growth is even faster (e.g.,$n^2e^{j\omega_0 n}$ for $k = 3$)

Stability

- Example: Evaluate the stability of the following two systems.

System 1: $y[n] = 2y[n-1] - y[n-2] + x[n]$

System 2: $y[n] = y[n-1] - y[n-2] + x[n]$

Stability

- Example: Evaluate the stability of the following two systems.

System 1: $y[n] = 2y[n-1] - y[n-2] + x[n]$

System 2: $y[n] = y[n-1] - y[n-2] + x[n]$

For system 1, the transfer function is

$$H_1(z) = \frac{1}{1 - 2z^{-1} + z^{-2}} = \frac{z^2}{z^2 - 2z + 1}$$

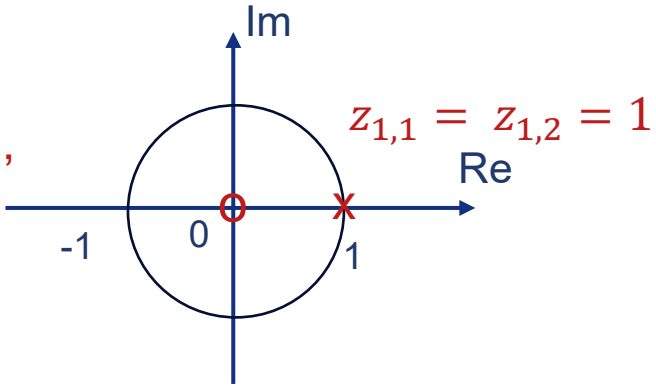
Hence, the poles are the solutions to $z^2 - 2z + 1 = 0$, which are given by

$$z_{1,1} = 1, z_{1,2} = 1$$

The pole-zero plot is shown on the right

The poles on the unit circle are duplicated.

Hence, the system is **unstable**



Stability

- Example: Evaluate the stability of the following two systems.

System 1: $y[n] = 2y[n-1] - y[n-2] + x[n]$

System 2: $y[n] = y[n-1] - y[n-2] + x[n]$

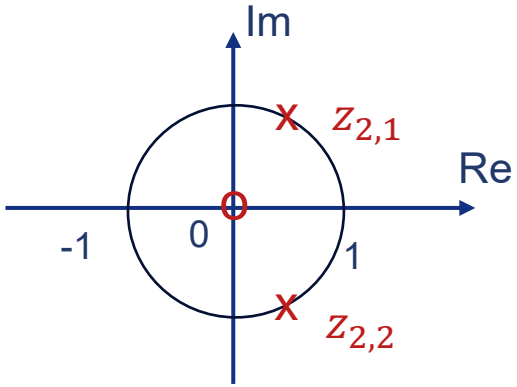
For system 2, the transfer function is

$$H_2(z) = \frac{1}{1 - z^{-1} + z^{-2}} = \frac{z^2}{z^2 - z + 1}$$

Hence, the poles are the solutions to $z^2 - z + 1 = 0$, which are given by

$$z_{2,1} = \frac{1+\sqrt{3}j}{2}, z_{2,2} = \frac{1-\sqrt{3}j}{2}$$

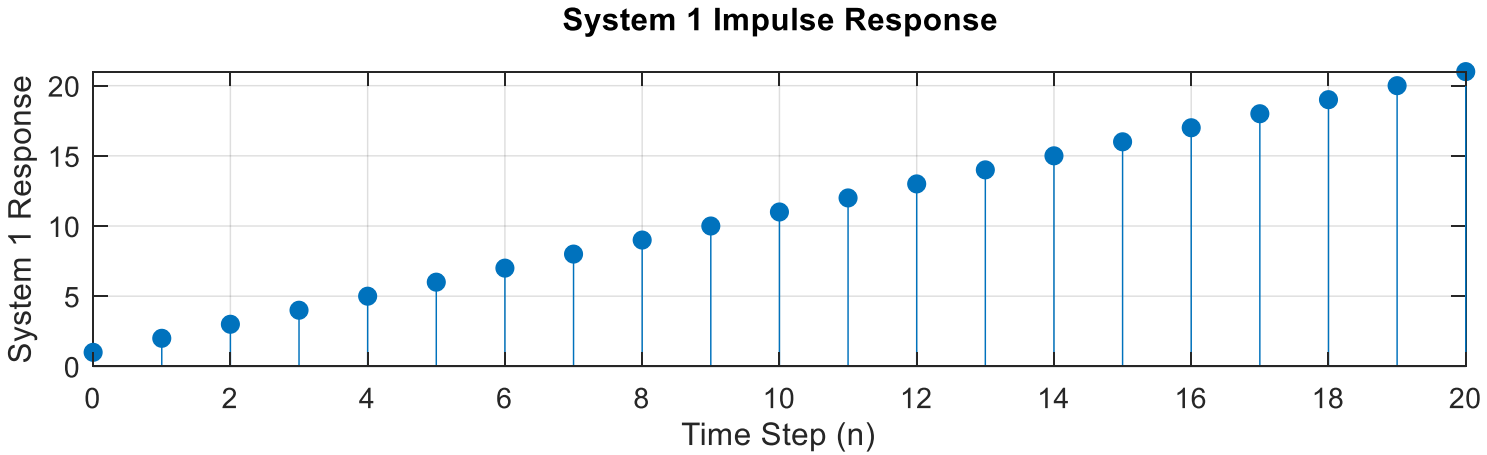
The pole-zero plot is shown on the right



The poles on the unit circle are distinct. Hence, the system is **marginally stable**

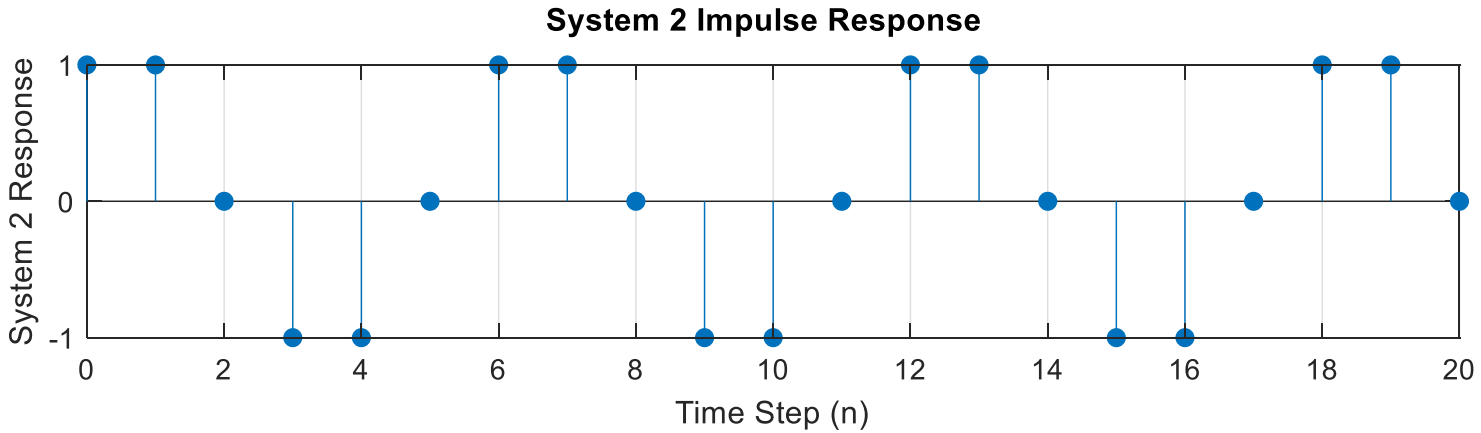
Stability

- We can confirm the stability of the two system by looking at their **impulse responses**
  1. The impulse response $h_1[n]$ of System 1 increases as n increases
     - This indicates that the system's output will grow to infinity given any input, which is defined as an unstable system



Stability

- We can confirm the stability of the two system by looking at their **impulse responses**
  2. The impulse response $h_2[n]$ of System 2 oscillates between 1 and -1
     - This indicates that the system's output is bounded but will never reach an equilibrium state, which is defined as a marginally stable system



Stability - Summary

Below is the procedure for evaluating the stability of a discrete system:

1. **Identify** the poles of the system.
2. **Check** if <u>all</u> poles are within the unit circle (|pole| < 1).
   - If **Yes**, the system is **Stable**.
   - If **No**, proceed to the next step.
3. **Check** if <u>any</u> pole is outside the unit circle (|pole| > 1)
   - If **Yes**, the system is **Unstable**.
   - If **No**, proceed to the next step.
4. **Check** for poles on the unit circle (|pole| = 1).
   - If **duplicate poles exist** on the unit circle, the system is **Unstable**.
   - If **no duplicated poles exist** on the unit circle, the system is **Marginally Stable**.

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